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WOMBLE BOND DICKINSON (US) LLP/Baidu USA LLC
ATTN: IP Docketing
P.O. Box 570489
Atlanta, GA 30357-0037

EXAMINER	
HILGENDORF, DALE W	

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3662	

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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/843,546	Applicant(s) Wu et al.	
	Examiner DALE W HILGENDORF	Art Unit 3662	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 6/17/2022.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☑ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 6/17/2022 is/are: a)☐ accepted or b)☑ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. Claims 1 thru 20 have been examined.

Drawings

3. The drawings are objected to because in Figure 5, the probability distribution 520 is associated with function p(x) and the probability distribution 530 is associated with function q(x). But in the specification P[0052] and P[0053, the "probability distribution 530 (P)" and the "probability distribution 520 (Q)" are recited. It is unclear if the P and Q are meant to be equated to the p(x) and q(x) of Figure 5, or if the P denotes a probability of success for 530 and Q denotes a probability of failure of 520. If they reference Figure 5, then the P and Q should refer to the other probability distribution. If they are the probability of success and failure, such explanation should be included in the specification (to avoid confusion with the functions p and q of Figure 5). Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as "amended." If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency.

Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

4. The drawings are objected to as failing to comply with 37 CFR 1.84(p)(5) because they do not include the following reference sign(s) mentioned in the description: From P[0068], reference numbers 808 and 809 are not in the drawings. From P[0069], reference numbers 813 and 814 are not in the drawings. Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

5. The drawings are objected to because in P[0090], Figure 14 and Figure 15 are recited, but here are not any Figure 14 or 15 provided. Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should

include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as "amended." If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Specification

6. The disclosure is objected to because of the following informalities: In P[0033] line 2, the reference to "expert demonstrations 112" should be "expert demonstrations 110" to agree with the description and the context of Figure 1.

Appropriate correction is required.

7. The disclosure is objected to because of the following informalities: In P[0048] line 5, the phrase "acceleration control 430, the traction" should be "acceleration control 430, and the traction".

Appropriate correction is required.

8. The use of the term BLUETOOTH (P[0064]), which is a trade name or a mark used in commerce, has been noted in this application. The term should be accompanied

by the generic terminology; furthermore the term should be capitalized wherever it appears or, where appropriate, include a proper symbol indicating use in commerce such as ™, SM , or ® following the term.

Although the use of trade names and marks used in commerce (i.e., trademarks, service marks, certification marks, and collective marks) are permissible in patent applications, the proprietary nature of the marks should be respected and every effort made to prevent their use in any manner which might adversely affect their validity as commercial marks.

Claim Interpretation

9. The following is a quotation of 35 U.S.C. 112(f):

> (f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

> An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

10. The claims in this application are given their broadest reasonable interpretation using the plain meaning of the claim language in light of the specification as it would be understood by one of ordinary skill in the art. The broadest reasonable interpretation of a claim element (also commonly referred to as a claim limitation) is limited by the description in the specification when 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is invoked.

As explained in MPEP § 2181, subsection I, claim limitations that meet the following three-prong test will be interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph:

(A) the claim limitation uses the term "means" or "step" or a term used as a substitute for "means" that is a generic placeholder (also called a nonce term or a non-structural term having no specific structural meaning) for performing the claimed function;

(B) the term "means" or "step" or the generic placeholder is modified by functional language, typically, but not always linked by the transition word "for" (e.g., "means for") or another linking word or phrase, such as "configured to" or "so that"; and

(C) the term "means" or "step" or the generic placeholder is not modified by sufficient structure, material, or acts for performing the claimed function.

Use of the word "means" (or "step") in a claim with functional language creates a rebuttable presumption that the claim limitation is to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites sufficient structure, material, or acts to entirely perform the recited function.

Absence of the word "means" (or "step") in a claim creates a rebuttable presumption that the claim limitation is not to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is not interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth

paragraph, is rebutted when the claim limitation recites function without reciting sufficient structure, material or acts to entirely perform the recited function.

Claim limitations in this application that use the word "means" (or "step") are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action. Conversely, claim limitations in this application that do not use the word "means" (or "step") are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action.

11. This application includes one or more claim limitations that do not use the word "means," but are nonetheless being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, because the claim limitation(s) uses a generic placeholder that is coupled with functional language without reciting sufficient structure to perform the recited function and the generic placeholder is not preceded by a structural modifier. Such claim limitation(s) is/are: generating and computing by a processing device in claim 1.

Because this/these claim limitation(s) is/are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, it/they is/are being interpreted to cover the corresponding structure described in the specification as performing the claimed function, and equivalents thereof. The claimed processing device is interpreted as a processor and memory based on the recited structure of claim 19.

If applicant does not intend to have this/these limitation(s) interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA

35 U.S.C. 112, sixth paragraph (e.g., by reciting sufficient structure to perform the claimed function); or (2) present a sufficient showing that the claim limitation(s) recite(s) sufficient structure to perform the claimed function so as to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph.

Claim Rejections - 35 USC § 101

12. 35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

13. Claims 1 thru 20 are rejected under 35 U.S.C. 101 because the claimed invention is directed to an abstract idea without significantly more.

Subject Matter Eligibility Criteria - Step 1:

14. Claim 1 is directed to a method (i.e., a process). Accordingly, claim 1 is within at least one of the four statutory categories. Claim 10 is directed to a non-transitory computer-readable storage medium (i.e., a machine). Accordingly, claim 16 is within at least one of the four statutory categories. Claim 19 is directed to a processing device (i.e., a machine). Accordingly, claim 19 is within at least one of the four statutory categories.

Subject Matter Eligibility Criteria - Alice/Mayo Test: Step 2A - Prong One:

15. Regarding Prong One of Step 2A of the Alice/Mayo test (which collectively includes the guidance in the January 7, 2019 Federal Register notice and the October 2019 update issued by the USPTO as now incorporated into the MPEP, as supported

by relevant case law), the claim limitations are to be analyzed to determine whether, under their broadest reasonable interpretation, they "recite" a judicial exception or in other words whether a judicial exception is "set forth" or "described" in the claims. MPEP 2106.04(II)(A)(1). An "abstract idea" judicial exception is subject matter that falls within at least one of the following groupings: a) certain methods of organizing human activity, b) mental processes, and/or c) mathematical concepts. MPEP 2106.04(a).

16. Independent claim 1 includes limitations that recite at least one abstract idea. Specifically, independent claim 1 recites:

A computer-implemented method comprising:

generating, by a processing device, a controlled trajectory of an autonomous driving vehicle (ADV) in a scenario, the controlled trajectory executable by the ADV to drive autonomously in the scenario;

receiving a set of data acquired in a plurality of driving demonstrations in the scenario;

identifying a distribution pattern of the set of data acquired, the distribution pattern indicating probabilities of driving trajectories in the scenario; and

computing, by the processing device, a similarity score based on comparisons between the controlled trajectory and the distribution pattern of the plurality of driving demonstrations.

17. The above underlined limitation constitutes "a mental process" because it is an observation/evaluation/judgment/analysis that can, at the currently claimed high level of generality, be practically performed in the human mind (e.g., with pen and paper). For

instance, a person could decide on a path for the vehicle to follow. This typically occurs when a person is driving a car, such as drive straight to the intersection, stop at the stop sign, and turn right when clear and you have the right of way. A person can use this knowledge even when they are not driving, just planning a trip. A person can then use acquired data (different options for the scenario) and look at the scenario data to identify patterns. A person can then determine which of the acquired scenarios are closest to their plan for the trip. See P[0022] of the present application which also disclose that the evaluations of the planning and control are based on human driving. The examiner notes that the claimed "controlled trajectory executable by the ADV to drive autonomously" is merely that it is able to be used and does not positively recite any control of the ADV. Accordingly, the claim recites at least one abstract idea.

18. Claims 10 and 19 recite the same limitations as the method steps of claim 1, with the addition of non-transitory machine-readable medium having instructions stored and executed by a processor to cause the processor to perform operations (claim 10), and a processor and a memory coupled to the processor storing instructions executed by the processor (claim 19). These claims also recite at least one abstract idea.

Subject Matter Eligibility Criteria - Alice/Mayo Test: Step 2A - Prong Two:

19. Regarding Prong Two of Step 2A of the Alice/Mayo test, it must be determined whether the claim as a whole integrates the abstract idea into a practical application. As noted at MPEP §2106.04(II)(A)(2), it must be determined whether any additional elements in the claim beyond the abstract idea integrate the exception into a practical application in a manner that imposes a meaningful limit on the judicial

exception. The courts have indicated that additional elements such as merely using a computer to implement an abstract idea, adding insignificant extra solution activity, or generally linking use of a judicial exception to a particular technological environment or field of use do not integrate a judicial exception into a "practical application." MPEP §2106.05(I)(A).

20. In the present case, the additional limitations beyond the above-noted at least one abstract idea recited in claim 1 are as follows (where the bolded portions are the "additional limitations" while the underlined portions continue to represent the at least one "abstract idea"):

A computer-implemented method comprising:

generating, by a **processing device** (using computers or machinery as mere tools to perform the abstract idea as noted below, see MPEP § 2106.05(f)), a controlled trajectory of an **autonomous driving vehicle (ADV)** (the machine is merely an object on which the method operates to perform the abstract idea as noted below, see MPEP § 2106.05(b)) in a scenario, the controlled trajectory executable by the ADV to drive autonomously in the scenario;

receiving a set of data acquired in a plurality of driving demonstrations in the scenario (extra-solution activity (data gathering) as noted below, see MPEP § 2106.05(g));

identifying a distribution pattern of the set of data acquired, the distribution pattern indicating probabilities of driving trajectories in the scenario; and

computing, by the **processing device** (using computers or machinery as mere tools to perform the abstract idea as noted below, see MPEP § 2106.05(f)), a similarity

score based on comparisons between the controlled trajectory and the distribution pattern of the plurality of driving demonstrations.

21. For the following reasons, the above-identified additional limitations, when considered as a whole with the limitations reciting the at least one abstract idea, do not integrate the above-noted at least one abstract idea into a practical application.

Regarding the additional limitation of a processing device, this limitation amounts to merely using a computer or other machinery as tools performing their typical functionality in conjunction with performing the above-noted at least one abstract idea (see MPEP § 2106.05(f)).

Regarding the additional limitation of an autonomous driving vehicle (ADV), this limitation amounts to merely providing the object on which the method operates to perform the above-noted at least one abstract idea (see MPEP § 2106.05(b)).

Regarding the additional limitation of receiving a set of data acquired in a plurality of driving demonstrations in the scenario, this additional limitation merely adds insignificant extra-solution activity (data gathering) to the at least one abstract idea in a manner that does not meaningfully limit the at least one abstract idea (see MPEP § 2106.05(g)).

Thus, taken alone, the additional elements do not integrate the at least one abstract idea into a practical application. Looking at the additional limitations as an ordered combination adds nothing that is not already present when looking at the elements taken individually. MPEP §2106.05(I)(A) and §2106.04(II)(A)(2).

For these reasons, claim 1 does not recite additional elements that integrate the judicial exception into a practical application. Accordingly, claim 1 is directed to at least one abstract idea.

22. Similarly, claim 10 recites, non-transitory machine-readable medium having instructions stored therein, which when executed by a processor, cause the processor to perform operations. The operations performed are the same as the method steps of claim 1, wherein the same analysis of the abstract idea also applies.

Regarding the additional limitations of non-transitory machine-readable medium having instructions stored therein, which when executed by a processor, cause the processor to perform operations, these limitations amount to merely using a computer or other machinery as tools performing their typical functionality in conjunction with performing the above-noted at least one abstract idea (see MPEP § 2106.05(f)).

Regarding the additional limitation of an autonomous driving vehicle (ADV), this limitation amounts to merely providing the object on which the method operates to perform the above-noted at least one abstract idea (see MPEP § 2106.05(b)).

Regarding the additional limitation of receiving a set of data acquired in a plurality of driving demonstrations in the scenario, this additional limitation merely adds insignificant extra-solution activity (data gathering) to the at least one abstract idea in a manner that does not meaningfully limit the at least one abstract idea (see MPEP § 2106.05(g)).

Thus, taken alone, the additional elements do not integrate the at least one abstract idea into a practical application. Looking at the additional limitations as an

ordered combination adds nothing that is not already present when looking at the elements taken individually. MPEP §2106.05(I)(A) and §2106.04(II)(A)(2).

For these reasons, claim 10 does not recite additional elements that integrate the judicial exception into a practical application. Accordingly, claim 10 is directed to at least one abstract idea.

23. Similarly, claim 19 recites, a processing device, comprising: a processor; and a memory coupled to the processor and storing instructions, which when executed by the processor, cause the processor to perform the operations that are the same as the method steps of claim 1, wherein the same analysis of the abstract idea also applies.

Regarding the additional limitation of a processing device, comprising: a processor; and a memory coupled to the processor and storing instructions, which when executed by the processor; this limitation amounts to merely using a computer or other machinery as tools performing their typical functionality in conjunction with performing the above-noted at least one abstract idea (see MPEP § 2106.05(f)).

Regarding the additional limitation of an autonomous driving vehicle (ADV), this limitation amounts to merely providing the object on which the method operates to perform the above-noted at least one abstract idea (see MPEP § 2106.05(b)).

Regarding the additional limitation of receive a set of data acquired in a plurality of driving demonstrations in the scenario, this additional limitation merely adds insignificant extra-solution activity (data gathering) to the at least one abstract idea in a manner that does not meaningfully limit the at least one abstract idea (see MPEP § 2106.05(g)).

Thus, taken alone, the additional elements do not integrate the at least one abstract idea into a practical application. Looking at the additional limitations as an ordered combination adds nothing that is not already present when looking at the elements taken individually. MPEP §2106.05(I)(A) and §2106.04(II)(A)(2).

For these reasons, claim 19 does not recite additional elements that integrate the judicial exception into a practical application. Accordingly, claim 19 is directed to at least one abstract idea.

Subject Matter Eligibility Criteria - Alice/Mayo Test: Step 2B:

24. Regarding Step 2B of the Alice/Mayo test, claims 1, 10, and 19 do not include additional elements (considered both individually and as an ordered combination) that are sufficient to amount to significantly more than the judicial exception for reasons the same as those discussed above with respect to determining that the claim does not integrate the abstract idea into a practical application.

Regarding claim 1, the additional limitation of a processing device, this limitation amounts to merely using a computer or other machinery as tools performing their typical functionality in conjunction with performing the above-noted at least one abstract idea (see MPEP § 2106.05(f)).

Regarding claim 10, the additional limitations of non-transitory machine-readable medium having instructions stored therein, which when executed by a processor, cause the processor to perform operations, these limitations amount to merely using a computer or other machinery as tools performing their typical functionality in conjunction with performing the above-noted at least one abstract idea (see MPEP § 2106.05(f)).

Regarding claim 19, the additional limitations of a processing device, comprising: a processor; and a memory coupled to the processor and storing instructions, which when executed by the processor, cause the processor to perform the operations; these limitations amount to merely using a computer or other machinery as tools performing their typical functionality in conjunction with performing the above-noted at least one abstract idea (see MPEP § 2106.05(f)).

Regarding the additional limitation of receiving a set of data acquired in a plurality of driving demonstrations in the scenario, this additional limitation has been reevaluated, and it has been determined that such limitations are not unconventional as they merely consist of data gathering which is recited at a high level of generality. See OIP Techs., Inc., v. Amazon.com, Inc., 788 F.3d 1359, 1363, 115 USPQ2d 1090, 1093 (Fed. Cir. 2015) (sending messages over a network); or buySAFE, Inc. v. Google, Inc., 765 F.3d 1350, 1355, 112 USPQ2d 1093, 1096 (Fed. Cir. 2014) (computer receives and sends information over a network). Further, adding a preliminary step of gathering data to a process that only recites identifying patterns and determining comparisons of the patterns (a mental process) does not add a meaningful limitation to the process of determining a similarity score of the scenario. See MPEP 2106.05(d)(II) and 2106.05(g).

25. The dependent claims 2 thru 9, 11 thru 18 and 20 do not provide additional elements or a practical application to become eligible under 35 U.S.C. 101. The dependent claims are directed to adjusting a parameter used for the generation of the trajectory and updating the trajectory based on previous computations (claims 2, 11, 20), generating candidate trajectories (claims 3, 12), calculation methods for the

similarity score (claims 4, 13) (also abstract idea of mathematical formulas), defining the scenario (claims 5, 6, 14, 15), performing mathematical calculations (claims 7, 16), defining the distribution pattern (claims 8, 17), and defining the source of the driving demonstrations (claims 9, 18). These limitations are extra-solution activity, or part of the abstract idea (mental activity and/or mathematical equations). They do not constitute a practical application of the abstract idea.

Claim Rejections - 35 USC § 103

26. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

27. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

28. Claim(s) 1, 2, 5, 6, 8 thru 11, 14, 15 and 17 thru 20 is/are rejected under 35 U.S.C. 103 as being unpatentable over He et al Patent Application Publication Number 2020/0363814 A1.

29. Regarding claims 1 and 10 He et al teach the claimed computer implemented method, the flow diagrams of Figures 9 and 14 implemented by processing logic (P[0070] and P[0080]), and the claimed machine readable medium having instructions

stored and executed by a processor to cause the processor to perform operation, "Storage device 1508 may include computer-accessible storage medium 1509 (also known as a machine-readable storage medium or a computer-readable medium) on which is stored one or more sets of instructions or software (e.g., module, unit, and/or logic 1528) embodying any one or more of the methodologies or functions described herein. Processing module/unit/logic 1528 may represent any of the components described above, such as, for example, machine learning engine 122 of FIG. 1, planning module 305, control module 306, and open space planning module 308 of FIG. 3A. Processing module/unit/logic 1528 may also reside, completely or at least partially, within memory 1503 and/or within processor 1501 during execution thereof by data processing system 1500, memory 1503 and processor 1501 also constituting machine-accessible storage media." P[0094], the operations comprising:

the claimed generating, by a processing device, a controlled trajectory of an autonomous driving vehicle (ADV) in a scenario, "At block 902, processing logic applies a reinforcement learning (RL) algorithm to an initial state of a planning trajectory based on the perceived environment to determine a list of controls for the ADV to advance to a list of trajectory states (e.g., next states) based on map and vehicle control information for the ADV." (P[0070] and Figure 9), the perceived environment and the map equate to the claimed scenario,

the claimed controlled trajectory is executable by the ADV to drive autonomously in the scenario, "Based on the planning and control data, control module 306 controls and drives the autonomous vehicle, by sending proper commands or signals to vehicle control system 111, according to a route or path

defined by the planning and control data. The planning and control data include sufficient information to drive the vehicle from a first point to a second point of a route or path using appropriate vehicle settings or driving parameters (e.g., throttle, braking, steering commands) at different points in time along the path or route." (P[0048] and Figure 3B);

the claimed receiving a set of data acquired in a plurality of driving demonstrations in the scenario, "at block 1401, processing logic generates a number of driving scenarios to train a RL agent. At block 1402, processing logic replays each of the driving scenarios to train the RL agent by, at block 1403, applying a reinforcement learning (RL) algorithm to an initial state of a driving scenario to determine a number of control actions from a number of discretized control action options, for the ADV to advance to a number of trajectory states from a number of discretized trajectory state options" (P[0080] and Figure 14);

the claimed identifying a distribution pattern of the set of data acquired, reinforcement learning (RL) refers to a type of machine learning technique that enables an agent (e.g., RL agent 803) to learn in an interactive environment (e.g., environment model 808) by trial and error using feedback from its actions and experiences, the machine learning (ML) relates to algorithms and statistical models to perform a specific task (here, to generate a driving trajectory) by relying on patterns and inferences P[0064],

the claimed distribution pattern indicating probabilities of driving trajectories in the scenario, "prediction module 303 may predict that the vehicle will more likely make a left turn or right turn" P[0044], "For each of the objects,

decision module 304 makes a decision regarding how to handle the object. For example, for a particular object (e.g., another vehicle in a crossing route) as well as its metadata describing the object (e.g., a speed, direction, turning angle), decision module 304 decides how to encounter the object (e.g., overtake, yield, stop, pass)." (P[0045] and Figure 3B), and "Decision module 304 and/or planning module 305 examine all of the possible routes to select and modify one of the most optimal routes in view of other data provided by other modules such as traffic conditions from localization module 301, driving environment perceived by perception module 302, and traffic condition predicted by prediction module 303." P[0046], the prediction of possible routes provided to the decision and planning equates to the claimed probabilities of driving trajectories in the scenario; and

the claimed computing a similarity score based on comparisons between the controlled trajectory and the distribution pattern of the plurality of driving demonstrations, "at block 1404, determining a reward prediction by the RL algorithm for each of the control actions, at block 1405, determining a judgment score (e.g., reward) for the trajectory states (e.g., a generated trajectory), and at block 1406, updating the RL agent based on the judgment score (e.g., reward)" (P[0080] and Figure 14), "the open space optimization model 421 can output a trajectory and controls (as a reference trajectory) to criteria evaluation 809 for criteria evaluation 809 to compare with the outputs of RL agent 803. Thus, RL agent 803 can learn from its actions and experiences in comparing to outputs of optimization model 421." P[0064], and "Criteria evaluation 809 can evaluate if the trajectory/controls generated by RL agent 803 satisfies a list of threshold criteria (or a threshold judgment score), if yes, the trajectory

and controls are output to outputs 811. If the judgment score is not satisfactory, criteria determiner 809 can trigger outputs 811 to output a second trajectory/second list of controls from open space optimization model 421. In one embodiment, the threshold criteria can be determined based on a feedback from open space optimization model 421." P[0064], the judgment score satisfying a threshold equates to the claimed similarity score.

He et al do not explicitly recite the claimed limitations are found in a single embodiment, but instead the claimed limitations are essentially taught in Figures 9 and 14. In reading the description of Figures 9 and 14 (P[0070] and P[0080] respectively), the steps of Figure 14 appears to be a continuation or expansion of the steps of Figure 9. Nevertheless, a person of ordinary skill in the art would understand that it would be obvious to combine the teachings of the methods of Figures 9 and 14 of He et al by making various modifications without departing from the broader spirit and scope of the disclosure P[0103]. It would have been obvious to a person having ordinary skill in the art before the effective filing date of the claimed invention to combine the process for generating a first trajectory from the trajectory states by maximizing the reward predictions to control the ADV autonomously according to the first trajectory of Figure 9 of He et al with the process for updating the RL agent based on the judgment score of Figure 14 of He et al in order to, with a reasonable expectation of success, improve control action outputs (He et al P[0076]).

30. Regarding claims 2 and 11 He et al teach the claimed method of claim 1 and the claimed machine readable medium of claim 10 (see above), further comprising:

the claimed adjusting at least one parameter in generating the controlled trajectory of the ADV in the scenario based on the similarity score, "at block 1406, updating the RL agent based on the judgment score" (P[0080] and Figure 14), and "Based on the reward prediction, actor NN 805 can be trained or updated to improve its control action outputs. RL agent 803 is coupled to environment model (state) 1109. Environment model (state) 1109 can generate a next trajectory state based on a control action. As such, RL agent 803 can interact with environment model (state) 1109 to iteratively output a number of trajectory states (e.g., $x_0 \ldots x_F$, a trajectory) and a number of control actions (e.g., e.g., $u_0 \ldots u_F$, controls and/or actions) for the RL agent to maneuver an ADV from an initial trajectory state (e.g., x_0) to a final trajectory state (e.g., x_F)." P{0076]; and

the claimed generating an updated controlled trajectory, "RL agent 803 can interact with environment model (state) 1109 to iteratively output a number of trajectory states (e.g., $x_0 \ldots x_F$, a trajectory) and a number of control actions (e.g., e.g., $u_0 \ldots u_F$, controls and/or actions) for the RL agent to maneuver an ADV from an initial trajectory state (e.g., x_0) to a final trajectory state (e.g., x_F)." P{0076], wherein

the claimed updated controlled trajectory corresponds to an updated similarity score higher than the similarity score computed previously, "Criteria evaluation 809 can evaluate if the trajectory/controls generated by RL agent 803 satisfies a list of threshold criteria (or a threshold judgment score), if yes, the trajectory and controls are output to outputs 811. If the judgment score is not satisfactory, criteria determiner 809 can trigger outputs 811 to output a second trajectory/second list of controls from open space optimization model 421. In one

embodiment, the threshold criteria can be determined based on a feedback from open space optimization model 421. For example, the open space optimization model 421 can output a trajectory and controls (as a reference trajectory) to criteria evaluation 809 for criteria evaluation 809 to compare with the outputs of RL agent 803. Thus, RL agent 803 can learn from its actions and experiences in comparing to outputs of optimization model 421." P[0064],

the claimed updated controlled trajectory is executable by the ADV to drive autonomously in the scenario, "Based on the planning and control data, control module 306 controls and drives the autonomous vehicle, by sending proper commands or signals to vehicle control system 111, according to a route or path defined by the planning and control data. The planning and control data include sufficient information to drive the vehicle from a first point to a second point of a route or path using appropriate vehicle settings or driving parameters (e.g., throttle, braking, steering commands) at different points in time along the path or route." (P[0048] and Figure 3B).

31. Regarding claims 5 and 14 He et al teach the claimed method of claim 1 and the claimed machine readable medium of claim 10 (see above), wherein the scenario comprises:

the claimed starting location and initial speed vector, "Routing module 307 is configured to provide one or more routes or paths from a starting point to a destination point." P[0046], and "planning module 305 plans a path or route for the autonomous vehicle, as well as driving parameters (e.g., distance, speed, and/or turning angle), using a reference line provided by routing module 307 as a basis" P[0047];

the claimed ending location and final speed vector, "Routing module 307 is configured to provide one or more routes or paths from a starting point to a destination point." P[0046], and "planning module 305 plans a path or route for the autonomous vehicle, as well as driving parameters (e.g., distance, speed, and/or turning angle), using a reference line provided by routing module 307 as a basis" P[0047]; and

the claimed static boundary, "Referring FIG. 5, in operation 501, processing logic extracts environment constraints from HD map, and moving obstacles constraints from prediction module to generate the open space optimization algorithm, and initializes matrices/vectors for the constraints of the optimization algorithm. In one embodiment, the optimization model can be previously generated and uploaded on the ADV (e.g., model 421 of FIG. 4, as part of models 313 of FIG. 3A). Constraints can include distances from the vehicle to the open road boundaries, dimensions of accessible roadways, road curbs, and one or more obstacles." P[0052], the road boundaries, dimensions of accessible roadways, and road curbs equate to the claimed static boundary.

32. Regarding claims 6 and 15 He et al teach the claimed scenario comprises at least one of a turn, a sequence change of directions, an overtake, a compliance to traffic signs or a timed attempt, "planning module 305 plans a path or route for the autonomous vehicle, as well as driving parameters (e.g., distance, speed, and/or turning angle)" P[0047] (claimed turn), and "Decision module 304 may make such decisions according to a set of rules such as traffic rules or driving rules 312" P[0045] (claimed compliance to traffic signs).

33. Regarding claims 8 and 17 He et al teach the claimed distribution pattern comprises an observed feature distribution of distances, speeds or acceleration, "RL agent 803 can learn from its actions and experiences in comparing to outputs of optimization model 421. As described above, open space optimization model 421 can include a model for vehicle dynamics of the ADV. Note, reinforcement learning (RL) refers to a type of machine learning technique that enables an agent (e.g., RL agent 803) to learn in an interactive environment (e.g., environment model 808) by trial and error using feedback from its actions and experiences. Machine learning (ML) relates to algorithms and statistical models to perform a specific task (here, to generate a driving trajectory) without using explicit instructions, but instead relies on patterns and inferences." P[0064], and "Server 103 may be a data analytics system to perform data analytics services for a variety of clients. In one embodiment, data analytics system 103 includes data collector 121 and machine learning engine 122. Data collector 121 collects driving statistics 123 from a variety of vehicles, either autonomous vehicles or regular vehicles driven by human drivers. Driving statistics 123 include information indicating the driving commands (e.g., throttle, brake, steering commands) issued and responses of the vehicles (e.g., speeds, accelerations, decelerations, directions) captured by sensors of the vehicles at different points in time. Driving statistics 123 may further include information describing the driving environments at different points in time, such as, for example, routes (including starting and destination locations), MPOIs, road conditions, weather conditions, etc." P[0037], the collected driving statistics equate to the claimed distances, speeds or acceleration, and the actions and experiences provided to the optimization model equate to the claimed observed feature distribution.

34. Regarding claims 9 and 18 He et al teach the claimed plurality of driving demonstrations comprises demonstrations performed in a simulation environment, "Environment model 1300 can model a simulated environment of an ADV (e.g., environment model (state) 1109 or environment model (reward) 1111 of FIG. 11) to interact with an RL agent to speed up reinforcement learning." P[0079].

35. Regarding claim 19 He et al teach the claimed processing device, "system 1500 includes processor 1501, memory 1503, and devices 1505-1508 connected via a bus or an interconnect 1510" (P[0087] and Figure 15), comprising:

the claimed processor, processor 1501 (Figure 15); and

the claimed memory coupled to the processor and storing instructions executed by the processor, "Processor 1501 may communicate with memory 1503, which in one embodiment can be implemented via multiple memory devices to provide for a given amount of system memory. Memory 1503 may include one or more volatile storage (or memory) devices such as random access memory (RAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), static RAM (SRAM), or other types of storage devices. Memory 1503 may store information including sequences of instructions that are executed by processor 1501, or any other device. For example, executable code and/or data of a variety of operating systems, device drivers, firmware (e.g., input output basic system or BIOS), and/or applications can be loaded in memory 1503 and executed by processor 1501." (P[0089] and Figure 15), cause the processor to

the claimed generate a controlled trajectory of an autonomous driving vehicle (ADV) in a scenario, "At block 902, processing logic applies a reinforcement learning (RL) algorithm to an initial state of a planning trajectory based on the perceived

environment to determine a list of controls for the ADV to advance to a list of trajectory states (e.g., next states) based on map and vehicle control information for the ADV." (P[0070] and Figure 9), the perceived environment and the map equate to the claimed scenario,

the claimed controlled trajectory is executable by the ADV to drive autonomously in the scenario, "Based on the planning and control data, control module 306 controls and drives the autonomous vehicle, by sending proper commands or signals to vehicle control system 111, according to a route or path defined by the planning and control data. The planning and control data include sufficient information to drive the vehicle from a first point to a second point of a route or path using appropriate vehicle settings or driving parameters (e.g., throttle, braking, steering commands) at different points in time along the path or route." (P[0048] and Figure 3B);

the claimed receive a set of data acquired in a plurality of driving demonstrations in the scenario, "at block 1401, processing logic generates a number of driving scenarios to train a RL agent. At block 1402, processing logic replays each of the driving scenarios to train the RL agent by, at block 1403, applying a reinforcement learning (RL) algorithm to an initial state of a driving scenario to determine a number of control actions from a number of discretized control action options, for the ADV to advance to a number of trajectory states from a number of discretized trajectory state options" (P[0080] and Figure 14);

the claimed identify a distribution pattern of the set of data acquired, reinforcement learning (RL) refers to a type of machine learning technique that enables

an agent (e.g., RL agent 803) to learn in an interactive environment (e.g., environment model 808) by trial and error using feedback from its actions and experiences, the machine learning (ML) relates to algorithms and statistical models to perform a specific task (here, to generate a driving trajectory) by relying on patterns and inferences P[0064],

the claimed distribution pattern indicating probabilities of driving trajectories in the scenario, "prediction module 303 may predict that the vehicle will more likely make a left turn or right turn" P[0044], "For each of the objects, decision module 304 makes a decision regarding how to handle the object. For example, for a particular object (e.g., another vehicle in a crossing route) as well as its metadata describing the object (e.g., a speed, direction, turning angle), decision module 304 decides how to encounter the object (e.g., overtake, yield, stop, pass)." (P[0045] and Figure 3B), and "Decision module 304 and/or planning module 305 examine all of the possible routes to select and modify one of the most optimal routes in view of other data provided by other modules such as traffic conditions from localization module 301, driving environment perceived by perception module 302, and traffic condition predicted by prediction module 303." P[0046], the prediction of possible routes provided to the decision and planning equates to the claimed probabilities of driving trajectories in the scenario; and

the claimed compute a similarity score based on comparisons between the controlled trajectory and the distribution pattern of the plurality of driving demonstrations, "at block 1404, determining a reward prediction by the RL algorithm for each of the control actions, at block 1405, determining a judgment score (e.g., reward)

for the trajectory states (e.g., a generated trajectory), and at block 1406, updating the RL agent based on the judgment score (e.g., reward)" (P[0080] and Figure 14), "the open space optimization model 421 can output a trajectory and controls (as a reference trajectory) to criteria evaluation 809 for criteria evaluation 809 to compare with the outputs of RL agent 803. Thus, RL agent 803 can learn from its actions and experiences in comparing to outputs of optimization model 421." P[0064], and "Criteria evaluation 809 can evaluate if the trajectory/controls generated by RL agent 803 satisfies a list of threshold criteria (or a threshold judgment score), if yes, the trajectory and controls are output to outputs 811. If the judgment score is not satisfactory, criteria determiner 809 can trigger outputs 811 to output a second trajectory/second list of controls from open space optimization model 421. In one embodiment, the threshold criteria can be determined based on a feedback from open space optimization model 421." P[0064], the judgment score satisfying a threshold equates to the claimed similarity score.

He et al do not explicitly recite the claimed limitations are found in a single embodiment, but instead the claimed limitations are essentially taught in Figures 9 and 14. In reading the description of Figures 9 and 14 (P[0070] and P[0080] respectively), the steps of Figure 14 appears to be a continuation or expansion of the steps of Figure 9. Nevertheless, a person of ordinary skill in the art would understand that it would be obvious to combine the teachings of the methods of Figures 9 and 14 of He et al by making various modifications without departing from the broader spirit and scope of the disclosure P[0103]. It would have been obvious to a person having ordinary skill in the art before the effective filing date of the claimed invention to combine the process for

generating a first trajectory from the trajectory states by maximizing the reward predictions to control the ADV autonomously according to the first trajectory of Figure 9 of He et al with the process for updating the RL agent based on the judgment score of Figure 14 of He et al in order to, with a reasonable expectation of success, improve control action outputs (He et al P[0076]).

36. Regarding claim 20 He et al teach the claimed processing device of claim 19 (see above), further cause the processor to:

the claimed adjust at least one parameter in generating the controlled trajectory of the ADV in the scenario based on the similarity score, "at block 1406, updating the RL agent based on the judgment score" (P[0080] and Figure 14), and "Based on the reward prediction, actor NN 805 can be trained or updated to improve its control action outputs. RL agent 803 is coupled to environment model (state) 1109. Environment model (state) 1109 can generate a next trajectory state based on a control action. As such, RL agent 803 can interact with environment model (state) 1109 to iteratively output a number of trajectory states (e.g., $x_0 \ldots x_F$, a trajectory) and a number of control actions (e.g., e.g., $u_0 \ldots u_F$, controls and/or actions) for the RL agent to maneuver an ADV from an initial trajectory state (e.g., x_0) to a final trajectory state (e.g., x_F)." P{0076]; and

the claimed generate an updated controlled trajectory, "RL agent 803 can interact with environment model (state) 1109 to iteratively output a number of trajectory states (e.g., $x_0 \ldots x_F$, a trajectory) and a number of control actions (e.g., e.g., $u_0 \ldots u_F$, controls and/or actions) for the RL agent to maneuver an ADV from an initial trajectory state (e.g., x_0) to a final trajectory state (e.g., x_F)." P{0076], wherein

the claimed updated controlled trajectory corresponds to an updated similarity score higher than the similarity score computed previously, "Criteria evaluation 809 can evaluate if the trajectory/controls generated by RL agent 803 satisfies a list of threshold criteria (or a threshold judgment score), if yes, the trajectory and controls are output to outputs 811. If the judgment score is not satisfactory, criteria determiner 809 can trigger outputs 811 to output a second trajectory/second list of controls from open space optimization model 421. In one embodiment, the threshold criteria can be determined based on a feedback from open space optimization model 421. For example, the open space optimization model 421 can output a trajectory and controls (as a reference trajectory) to criteria evaluation 809 for criteria evaluation 809 to compare with the outputs of RL agent 803. Thus, RL agent 803 can learn from its actions and experiences in comparing to outputs of optimization model 421." P[0064],

the claimed updated controlled trajectory is executable by the ADV to drive autonomously in the scenario, "Based on the planning and control data, control module 306 controls and drives the autonomous vehicle, by sending proper commands or signals to vehicle control system 111, according to a route or path defined by the planning and control data. The planning and control data include sufficient information to drive the vehicle from a first point to a second point of a route or path using appropriate vehicle settings or driving parameters (e.g., throttle, braking, steering commands) at different points in time along the path or route." (P[0048] and Figure 3B).

37. Claim(s) 3, 4, 12 and 13 is/are rejected under 35 U.S.C. 103 as being unpatentable over He et al Patent Application Publication Number 2020/0363814 A1 in view of Kusari Patent Application Publication Number 2020/0142420 A1.

38. Regarding claims 3 and 12 He et al teach the claimed method of claim 1 and the claimed machine readable medium of claim 10 (see above). He et al do not teach the claimed generating the controlled trajectory of the ADV scenario comprises generating a plurality of candidate trajectories subject to a probability distribution pattern. He et al do rely on patterns of the actions and experiences of an optimization model (learned as reference trajectories) to provide trajectory control of the ADV (P[0063] and p[0064]). A probability distribution of the pattern would provide a more accurate range of actions and experiences used to control the ADV.

Kusari teaches generating a plurality of candidate trajectories subject to a probability distribution pattern, "expert states 302 and expert actions 304 can be enumerated with respect to vehicle trajectories and probability distributions of cumulative rewards can be calculated for the enumerated expert states 302 and expert actions 304" P[0032], and "A mathematical model of vehicle 110 can be determined empirically by recording a plurality of controller instructions and vehicle 110 trajectories corresponding to the controller instructions and determining mathematical relationships between vehicle 110 trajectories and controller instructions." P[0023], the expert states and actions equate to the previously claimed driving demonstrations, and the vehicle trajectories and controller instructions equate to the claimed plurality of candidate trajectories. It would have been obvious to a person having ordinary skill in the art before the effective filing date of the claimed invention to combine the reinforcement

learning of trajectory planning for autonomous vehicles of He et al with the expert states and actions used for probability distribution when determining vehicle trajectories of Kusari in order to, with a reasonable expectation of success, achieve safe and efficient traversal of a route (Kusari P[0017]).

39. Regarding claims 4 and 13 He et al do not teach the claimed computing of the similarity score comprises at least one of calculating a Kullback-Leibler divergence or evaluating a probability density of a predicted value of a probability distribution function of the distribution pattern. Kusari teaches, "Kullback-Lieber divergence is a measure in information loss corresponding to a difference between the probability distributions of the posterior and the prior reward functions." P[0037] (Kusari has misspelled "Leibler"), and "The information regarding cumulative reward functions v_t can be stored at prior processing block 310 as probability density distributions." P[0031]. It would have been obvious to a person having ordinary skill in the art before the effective filing date of the claimed invention to combine the reinforcement learning of trajectory planning for autonomous vehicles of He et al with the expert states and actions used for probability distribution when determining vehicle trajectories using KL divergence or probability density of Kusari in order to, with a reasonable expectation of success, achieve safe and efficient traversal of a route (Kusari P[0017]).

Allowable Subject Matter

40. Claims 7 and 16 would be allowable if rewritten to overcome the rejection(s) under 35 U.S.C. 101, set forth in this Office action and to include all of the limitations of the base claim and any intervening claims.

Related Art

41. The examiner points to Li et al PGPub 2018/0374359 A1 as related art, but not relied upon for any rejection. Li et al is directed to training a deep neural network and evaluating a prediction method of autonomous driving (Figures 7 and 8). More specifically, Figure 8 is directed to "operation 801, processing logic receives a predicted trajectory (e.g., unknown predicted trajectory), which was generated using a prediction method or algorithm based on perception data perceiving a driving environment surrounding an ADV. For at least some of the trajectory points of the predicted trajectory, in operation 802, processing logic extracts a set of features from the selected trajectory points. In operation 803, processing logic applies a predetermined DNN model to the features to categorize the features and generate a similarity score. The similarity score represents the similarity between the predicted trajectory and one or more prior actual trajectories. In operation 804, processing logic determines a performance or accuracy of the prediction method or algorithm based on the similarity score." P0065].

Conclusion

42. Any inquiry concerning this communication or earlier communications from the examiner should be directed to DALE W HILGENDORF whose telephone number is (571)272-9635. The examiner can normally be reached Monday - Friday 9-5:30.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jelani Smith can be reached on 571-270-3969. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/DALE W HILGENDORF/
Primary Examiner, Art Unit 3662